

Supplementary Information

A Multi-Vendor Corpus of Reference Low-Dose CT Reconstructions with Calibrated Per-Pixel Uncertainty and Signal-Equivalence Credentials

This supplement records the configuration needed to regenerate the corpus bit-for-bit (within IEEE-754 tolerance) from the deposited Apache-2.0 RunBundle. **Table S1** lists the reconstruction and training hyperparameters; **Table S2** lists the determinism settings and pinned software versions; **Table S3** discloses training compute. Entries marked **[TODO:]** are fixed when the Phase-3 models are trained and the corpus is deposited.

S1 Hyperparameters

S2 Reproducibility: determinism and software pins

S3 Training-compute disclosure

Group	Hyperparameter	Value
<i>Unrolled reconstruction</i>		
	Iterations K	10
	Warm start	TV-regularized FBP (TV-FBP)
	Data-consistency step size τ	learned scalar, init $0.5/\ \mathcal{R}\ _2^2$
	Reconstruction loss	mean absolute error (L1) on HU
	Forward / adjoint operator	<code>pwm_core ct_radon</code> (version in Table S2)
	Projection geometry ($n_{\text{views}}, n_{\text{dets}}$)	[TODO: geometry]
	FBP filter	[TODO: filter (e.g. Ram-Lak / Hann)]
<i>Denoiser f_θ (weights shared across iterations)</i>		
	Architecture	U-Net, 4 encoder / 4 decoder stages
	Channels per stage	(32, 64, 128, 256)
	Activation / normalisation	GELU / GroupNorm
	Upsampling	bilinear
	Parameters per ensemble member	≈ 4 M
<i>Optimisation</i>		
	Optimiser	AdamW
	Peak learning rate	[TODO: lr]
	Weight decay	[TODO: wd]
	LR schedule	cosine, 5% linear warmup
	Training length (epochs / steps)	[TODO: epochs]
	Batch size	4
	Input	full 512×512 axial slice (2D)
<i>Training data</i>		
	Dose levels r	{0.10, 0.25, 0.50}
	Per-minibatch dose-mix weights	(0.4, 0.4, 0.2) for $r = (0.10, 0.25, 0.50)$
	Augmentation	flip $p=0.5$; rotation $[-5^\circ, 5^\circ]$; window-level shift $[-50, 50]$ HU
	Train / val / test sizes (patients)	[TODO: N / N / N]
<i>Ensemble</i>		
	Members M	5
	Member seeds	{42, 43, 44, 45, 46}

Table S1: Reconstruction, denoiser, optimisation, data, and ensemble hyperparameters. Fixed values mirror the Methods; **[TODO:]** entries are set at training time.

Group	Setting / package	Value
<i>Seeding and determinism</i>		
	Seeded RNG streams	Python <code>random</code> , NumPy, PyTorch (CPU + CUDA), DataLoader workers
	<code>cuda.deterministic</code>	True
	<code>cuda.benchmark</code>	False
	<code>torch.use_deterministic_algorithms</code>	[TODO: True/False + any opt-outs]
	Per-member seed $\rightarrow$ weights + data order	{42, ..., 46} (Table S1)
<i>Pinned software versions</i>		
	Python	[TODO: 3.x.y]
	PyTorch	[TODO: x.y.z]
	CUDA / cuDNN	[TODO: x.y / x.y]
	NumPy	[TODO: x.y.z]
	SciPy	[TODO: x.y.z]
	<code>pwm_core (ct_radon)</code>	[TODO: commit / version]
	<code>pwm_dose_equivalence</code>	0.2.2
	<code>nnDetection</code> (frozen task detector)	[TODO: release / commit]
	<code>nibabel</code>	[TODO: x.y.z]
	Docker base image digest	[TODO: sha256:...]
<i>Reproduction tolerance</i>		
	Bit-identical within	IEEE-754 FP tolerance on equivalent hardware
	Framework hash (credentials)	sha256:b366f51c... (<code>pwm_dose_equivalence</code>)

Table S2: Determinism settings, pinned software versions, and the framework hash credentials are issued under. These plus the RunBundle permit bit-reproduction of every record.

Quantity	Value
GPU model	NVIDIA A100 (single)
GPU-hours per ensemble member	[TODO: h]
Ensemble members trained	5 (reference) + cross-vendor leave-one-out variants
Total training GPU-hours (all models)	[TODO: h]
Inference time per slice (5-model ensemble)	[TODO: s on one A100]
Peak GPU memory (training)	[TODO: GB]
Estimated energy / CO ₂ e	[TODO: kWh / kg (optional)]

Table S3: Training-compute disclosure for the released models.